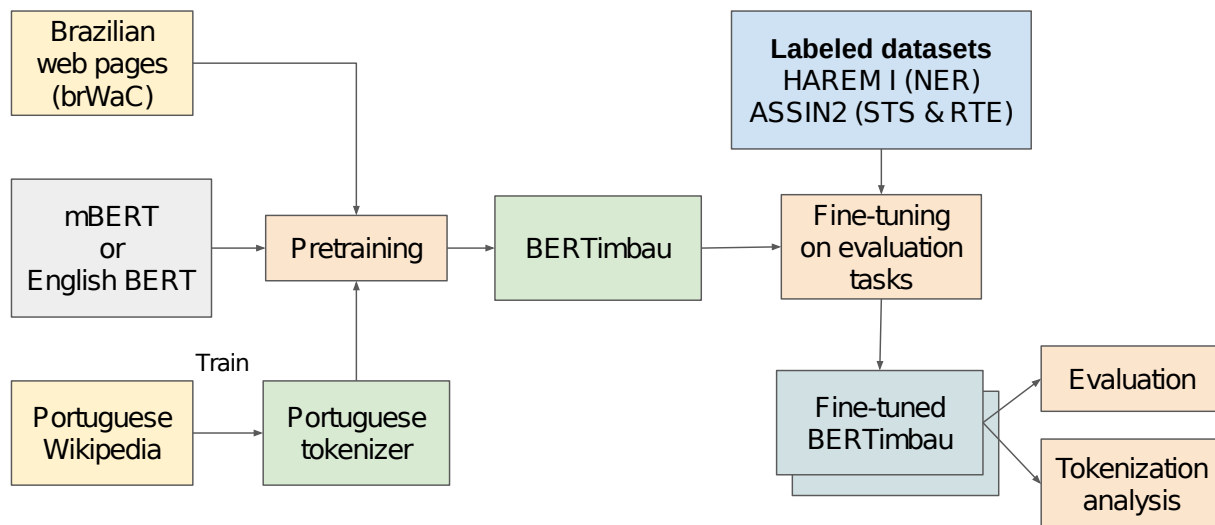


Graphical Abstract

BERT models for Brazilian Portuguese: pretraining, evaluation and tokenization analysis

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Highlights

BERT models for Brazilian Portuguese: pretraining, evaluation and tokenization analysis

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- Release of pretrained BERT models for Brazilian Portuguese trained on the brWaC corpus
- State-of-the-art performance on three Portuguese NLP tasks: Sentence Textual Similarity (ASSIN2), Recognizing Textual Entailment (ASSIN2) and Named Entity Recognition (HAREM I)
- Tokenization analysis reveals strong correlation between task performance and subword splits and opens future research directions

BERT models for Brazilian Portuguese: pretraining, evaluation and tokenization analysis

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ARTICLE INFO

Keywords:

Language model
BERT
Sentence textual similarity
Recognizing textual entailment
Named entity recognition

ABSTRACT

Recent advances in language representation using neural networks have made it viable to transfer the learned internal states of large pretrained language models (LMs) to downstream natural language processing (NLP) tasks. This transfer learning approach improves the overall performance on many tasks and is highly beneficial when labeled data is scarce, making pretrained LMs valuable resources specially for languages with few annotated training examples. In this work, we train BERT (Bidirectional Encoder Representations from Transformers) models for Brazilian Portuguese, which we nickname BERTimbau. We evaluate our models on three downstream NLP tasks: sentence textual similarity, recognizing textual entailment, and named entity recognition. Our models improve the state-of-the-art in all of these tasks, outperforming Multilingual BERT and confirming the effectiveness of large pretrained LMs for Portuguese. We release our models to the community hoping to provide strong baselines for future NLP research: <https://github.com/neuralmind-ai/portuguese-bert>.

1. Introduction

Transfer learning, where a model is first trained on a source task and then fine-tuned on tasks of interest, has changed the landscape of natural language processing (NLP) applications over the last years. The strategy of fine-tuning a large pretrained language model (LM) has been largely adopted and has achieved state-of-the-art performance on a variety of NLP tasks [11, 35, 37, 58]. Aside from bringing performance improvements, transfer learning reduces the amount of labeled data needed for supervised learning on downstream tasks [15, 32].

Pretraining these large language models, however, requires huge amounts of unlabeled data and computational resources, with reports of models being trained using thousands of GPUs or TPUs and hundreds of GBs of raw textual data [25, 37]. This resource barrier has limited the availability of these models, early on, to English, Chinese and multilingual models.

BERT [11], which uses the Transformer architecture [53], among its derived models, such as RoBERTa [25] and Albert [23], is one of the most adopted models. Despite having a multilingual BERT¹ model (mBERT) trained on 104 languages, much effort has been devoted on pretraining monolingual BERT and BERT-derived models on single languages, such as French [26], Dutch [10, 54], Spanish [4], Italian [34], and others [2, 20, 28]. Even though it is infeasible to train monolingual models for every language, these works are motivated by the superior performance on downstream tasks and resource efficiency of monolingual models compared to mBERT [16].

Large pretrained LMs can be valuable assets especially for languages that have few annotated resources but abundant unlabeled data, such as Portuguese. With that in mind, we train BERT models for Brazilian Portuguese — which we nickname BERTimbau — using data from brWaC [55], a large and diverse corpus of web pages. We evaluate our models on three NLP tasks: sentence textual similarity, recognizing textual entailment, and named entity recognition. We also compare the BERTimbau's and mBERT's tokenizers on these tasks and analyze their implications on the tasks' performances. BERTimbau improves the state-of-the-art on these tasks over multilingual models and previous monolingual approaches, confirming the effectiveness of large pretrained LMs for Portuguese. This work extends our previous work [48], in which we perform the pretraining and evaluation of BERTimbau models. In this work, we conduct a tokenization analysis of BERTimbau and mBERT tokenizers and their impacts on the downstream

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¹<https://github.com/google-research/bert/blob/master/multilingual.md>

tasks performance, which reveals a correlation between segmentation of words and task performance. Moreover, we improve methodology details, specially regarding vocabulary generation, pretraining and finetuning stages, more detailed description of evaluation datasets and tasks, and discussion of related and future work. We make BERTimbau models available to the community² on open-source libraries to provide strong baselines for future research and empower transfer learning on NLP applications in scenarios of limited labeled data or insufficient data to train a model from scratch.

The paper is organized as follows: in Section 2, we present the related work. In Section 3, we briefly describe BERTimbau architecture and the pretraining procedure, such as the pretraining data, the vocabulary generation, and pretraining objectives. In Section 4, we describe the downstream tasks and datasets used to evaluate our models and the evaluation procedures. In Section 5, we describe our experiments and present and analyze our results. Then, in Section 6, we perform tokenization comparison and analysis of its impacts on the evaluation tasks. Lastly, we make our final remarks in Section 7.

2. Related Work

Transfer learning is a widespread technique to reduce training data requirements and improve performance of machine learning models [30]. Sequential inductive transfer learning consists of training a model on a source dataset and task and then transferring the learned general-purpose features to a new target dataset and task [59]. This transfer is generally done by initializing all or part of the target model’s parameters with the pretrained weights of the source model. These pretrained weights can be kept frozen – and used in a feature extraction approach – or be fine-tuned along with new parameters of the target model when training on the final task [33].

Before the advent of pretrained LMs, most of the transfer learnings were made by first learning vector representations for predefined vocabularies in an unsupervised manner on large corpora and later reusing these representations as input to models that were trained from scratch. Classic word embeddings [27, 31] consists of static non-contextualized word-level representations that capture semantic and syntactic features. More recently, contextual embeddings, such as ELMo [32] and Flair Embeddings [1], leverage the internal states of language models to extract richer word representations in context.

Deep transfer learning techniques for NLP emerged by successfully fine-tuning large pretrained LMs with general-purpose architectures, such as the Transformer [53], replacing task-specific models. Language modeling pretraining is shown to resemble a multitask objective that allows zero-shot learning on many tasks [36]. This pretraining stage benefits from diverse texts and can be further improved by additional pretraining with unlabeled data of downstream tasks’ domains [14].

Pretrained LMs can be monolingual or multilingual. Initial works focused primarily on English monolingual LMs [15, 35], while BERT [11] also released a multilingual model (mBERT) that is trained on 104 languages. While multilingual models can be applied to any compatible language and allow cross-lingual transfer, they suffer from the *curse of multilinguality* [8, 7]: as more languages are added to a fixed-capacity model, the overall monolingual and transfer performances degrade, specially for low-resource languages that are underrepresented in the training data and vocabulary. This effect can be mitigated by increasing the model capacity, which results in larger compute resource requirements.

Similar to BERTimbau, much effort has been devoted to pretraining monolingual BERT and BERT-derived models for other languages, such as French [26], Dutch [10, 54], Spanish [4], Italian [34], and many others [2, 20, 28]. These works usually follow a common recipe and hyperparameters from the original works for the model pretraining, since the high cost of pretraining hinders much experimentation and optimization. The main differences between BERTimbau and the other works are the pretraining corpus and choices of model initialization, tokenizer, vocabulary generation, evaluation tasks and datasets. For an in-depth comparison of the works, we refer readers to AMMUS [16], a monolingual LMs survey.

For the Portuguese language, recent works mainly explored and compared contextual embedding techniques. ELMo and Flair Embeddings trained on a large Portuguese corpora achieve state-of-the-art results on IberLEF 2019 and HAREM named entity recognition (NER) tasks [5, 44, 9]. A comparison of ELMo and multilingual BERT using a contextual embeddings setup shows superior performance of Portuguese ELMo on semantic textual similarity task when no fine-tuning is used [41]. Regarding pretrained LMs for specific domains, BioBERTpt [45], a finetuned mBERT on clinical and biomedical texts in Portuguese, achieves state-of-the-art on SemClinBr NER tasks. BERTaú [12], a

²<https://github.com/neuralmind-ai/portuguese-bert>

BERT model trained from scratch on Portuguese texts of the banking domain, achieves state-of-the-art performance on private in-domain information retrieval, sentiment analysis and NER tasks.

3. BERTimbau

In this section, we present a high-level overview of BERT’s architecture and describe the procedures to pretrain BERT models for Brazilian Portuguese.

3.1. BERT overview

BERT (Bidirectional Encoder Representation from Transformers) [11] is a language model based on the Transformer Encoder [53] architecture. BERT’s main contribution is establishing a framework to pretrain deep bidirectional word representations that are jointly conditioned on both left and right contexts, in contrast to preceding works on language modeling that employ unidirectional LMs. For example, OpenAI GPT [35] pretrains a Transformer Decoder using left-to-right language modeling, and ELMo [32] concatenates representations from independent left-to-right and right-to-left language models. BERT demonstrates that bidirectionality is important for both sentence-level and token-level tasks by improving the state-of-the-art on several benchmarks. This unidirectionality limitation is overcome by using a modified language modeling objective called Masked Language Modeling, that resembles a denoising objective and which we detail later in this section.

A Transformer Encoder consists of a stack of L identical layers $h_i : \mathbb{R}^{N \times H} \mapsto \mathbb{R}^{N \times H}$ of hidden dimension H that are applied consecutively:

$$h(x) = h_L(h_{L-1}(\dots h_2(h_1(x)) \dots)), \quad (1)$$

where N is the length of the input sequence in tokens. Each layer contains a multi-head self-attention layer and a feedforward layer, which is a MLP network with two layers.

BERT can receive as input a single sentence or a sentence pair.³ An input sequence is generated by packing the input sentences using two special tokens, [CLS] and [SEP]. A single sentence of tokens $(x_1, \dots, x_N)$ is represented as

$$[\text{CLS}] \ x_1 \ \dots \ x_N \ [\text{SEP}], \quad (2)$$

and a sentence pair $(x_1, \dots, x_N)$ and $(y_1, \dots, y_M)$, as

$$[\text{CLS}] \ x_1 \ \dots \ x_N \ [\text{SEP}] \ y_1 \ \dots \ y_M \ [\text{SEP}], \quad (3)$$

where x_i and y_i are tokens of a vocabulary $\mathcal{V}$ of size V and N, M are the number of tokens in each sentence. The [SEP] token is simply a separator to mark the end of a sentence.

For the purposes of this work, BERT can be seen as a black box model that embeds and maps a sequence of tokens $\mathbf{x}$ into a sequence of encoded token representations:

$$(x_1, \dots, x_N) \mapsto (\mathbf{c}, \mathbf{T}_1, \dots, \mathbf{T}_N) \quad (4)$$

where $\mathbf{T}_i \in \mathbb{R}^H$ is the encoded representation of the i -th token x_i in the sequence $\mathbf{x}$, and $\mathbf{c} \in \mathbb{R}^H$ is the encoded output of the [CLS] token, which is used as an aggregate representation of the entire sequence for sequence-level tasks. To apply BERT to a task of interest, the representations $\mathbf{c}$ or $\mathbf{T}_i$ are used as inputs to a task specific model, which can be as simple as a linear transformation. We refer readers to Vaswani et al. [53] and Devlin et al. [11] for further details on Transformer and BERT architectures.

BERT usage in downstream tasks is composed of two stages: pretraining and fine-tuning. In the pretraining stage, the model is trained from scratch on self-supervised tasks to learn useful representations $\mathbf{c}$ and $\mathbf{T}_i$. This stage is computationally intensive and has to be performed only once. In the fine-tuning stage, a task specific model is attached to the pretrained BERT and the whole model is further trained on the task of interest.

³We borrow the notation of the original work: a “sentence” can be any arbitrary contiguous text span, rather than a linguistic sentence. A “sequence”, in its turn, refers to the input token sequence, which can be composed of one or two sentences.

In the pretraining stage, BERT is trained on two self-supervised tasks: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). Each pretraining example is generated by concatenating two sentences A, with tokens $(a_1, \dots, a_N)$ and B, with tokens $(b_1, \dots, b_M)$, as shown by Eq. 3, where N and M are the sentences' lengths. Given a sentence A from the corpus, in 50% of the time the sentence B is the sentence that follows sentence A, forming a contiguous piece of text, and 50% of the time B is a random sentence sampled from a distinct document of the corpus. This choice defines the ground truth label for the NSP task

$$y_{\text{NSP}} = \mathbb{1}(\text{B is the continuation of A}) \quad (5)$$

where $\mathbb{1}(\cdot)$ is the indicator function.

For MLM, each example is then corrupted by first selecting a random set of positions (integers from 1 to $N = |x|$) $\mathbf{m} = [m_1, \dots, m_K]$, $K = \lceil 0.15N \rceil$. The tokens in each of these positions are then replaced by 1 of 3 options: a special [MASK] token with 80% probability, a random token from the vocabulary with 10% probability or, otherwise, keeping the original token. In this selection step we use whole word masking: if a token from a word composed of multiple subword units is chosen to be corrupted, all other subword units are also corrupted. The original tokens from the positions m_i are saved and serve as labels for the MLM task

$$y_{\text{MLM}} = [x_{m_1}^*, \dots, x_{m_K}^*]. \quad (6)$$

The final pretraining example can be represented as a tuple $(x^{\text{corrupted}}, \mathbf{m}, y_{\text{MLM}}, y_{\text{NSP}})$.

The corrupted sequences $x^{\text{corrupted}}$ are used as inputs to BERT and the output encoded representations are used as inputs for the pretraining tasks' heads that are attached during this pretraining stage: the NSP head uses the $\mathbf{c}$ vector and the MLM head uses the $\mathbf{T}_i$ vectors of the masked positions. Both heads perform a classification task and the complete model is optimized by minimizing the sum of the cross-entropy losses of the two tasks.

3.2. Vocabulary generation

BERT uses WordPiece [46], a subword tokenization technique that is not open-source. As an alternative, we generate a cased Portuguese vocabulary of 30,000 subword units using the SentencePiece library [19] with the BPE algorithm [47] and 2,000,000 random sentences from Portuguese Wikipedia, which contains close to 1 million articles. The resulting vocabulary is then converted to WordPiece format for compatibility with original BERT code.

To convert the generated SentencePiece vocabulary to WordPiece format, we follow BERT's tokenization rules. Firstly, all BERT special tokens ([CLS], [MASK], [SEP], and [UNK]) and all punctuation characters of mBERT's vocabulary are added to the Portuguese vocabulary. Then, since BERT splits the text at whitespace and punctuation prior to applying WordPiece tokenization in the resulting chunks, each SentencePiece token that contains punctuation characters is split at these characters, the punctuations are removed and the resulting subword units are added to the vocabulary.⁴ Finally, subword units that do not start with SentencePiece's meta character "▁" are prefixed with "##" and the "▁" symbol is removed from the remaining tokens.

3.3. Pretraining data

For pretraining data, we use the brWaC [55] corpus (Brazilian Web as Corpus), a crawl of Brazilian webpages which contains 2.68 billion tokens from 3.53 million documents and is the largest open Portuguese corpus to date. On top of its size, brWaC is composed of whole documents and its methodology ensures high domain diversity and content quality, which are desirable features for BERT pretraining.

We use only the document body (ignoring the titles) and we apply a single post-processing step on the data to remove *mojibakes*⁵ and remnant HTML tags using the *ftfy* library [49]. The final processed corpus has 17.5GB of raw text. We split the corpus into chunks of 50MB and generate pretraining examples independently for each file as described in 3.1, with a duplication factor of 10. That is, we run example generation 10 times for each 50MB file, producing distinct sentence pairs for NSP task and token masks for MLM task. For maximum sequence length of 128 tokens, a total of 4.29×10^8 examples are generated, and, for maximum length 512, a total of 1.58×10^8 examples.

⁴Splitting at punctuation implies no subword unit can contain both punctuation and non-punctuation characters. Also, there cannot be two punctuation characters inside a single subword unit.

⁵Mojibake is a kind of text corruption that occurs when strings are decoded using the incorrect character encoding. For example, the word "codificação" becomes "codificaÃ§Ã£o" when encoded in UTF-8 and decoded using ISO-8859-1.

4. Evaluation

Once pretrained, we evaluate our models on 3 downstream NLP tasks: Sentence Textual Similarity (STS), Recognizing Textual Entailment (RTE), and Named Entity Recognition (NER). These tasks were chosen as to assess both sentence- and token-level tasks and because of the availability of labeled datasets and comparable strong baselines. To evaluate BERTimbau on downstream tasks, we remove the MLM and NSP classification heads used during pretraining stage and attach a relevant head required for each task. We then fine-tune our models on each task or pair of tasks. Similar to pretraining, sentence-level tasks are performed on the encoded representation of the [CLS] special token, $\mathbf{c}$, and token-level tasks use the encoded representation of each relevant token, $\mathbf{T}_i$.

4.1. Sentence Textual Similarity and Recognizing Textual Entailment

Sentence Textual Similarity (STS) is a regression task that measures the degree of semantic equivalence between two sentences in a numeric scale. Recognizing Textual Entailment (RTE), also known as Natural Language Inference (NLI), is a classification task of predicting if a given premise sentence entails a hypothesis sentence.

We use the ASSIN2 dataset [38], a shared task with 10,000 sentence pairs with STS and RTE annotations. The dataset is composed of 6500 train, 500 validation and 3000 test examples.

STS labels are continuous values in a scale of 1 to 5, where a pair of sentences with completely different meanings have label value of 1 and semantically equivalent sentences have value of 5. STS performance is evaluated using Pearson's Correlation as primary metric and Mean Squared Error (MSE) as secondary metric.

RTE labels are simply Entailment and None. RTE performance is evaluated using macro F1-score as primary metric and accuracy as secondary metric. Examples from ASSIN2 dataset can be seen in Table 1, which contains sentence pairs and their corresponding gold labels for both tasks.

Given an example with a premise sentence and a hypothesis sentence, we concatenate the two sentences as of Eq. 3 and feed the sequence into BERTimbau. We attach two independent linear layers on top of BERTimbau in a multitask scheme to predict the STS relatedness score and the RTE classification. We train using MSE loss for STS and cross-entropy loss for RTE. The final loss is the sum of both losses with equal weight.

4.2. Named Entity Recognition

Named Entity Recognition (NER) is the task of identifying text spans that mention named entities (NEs) and classifying them into predefined categories, such as person, organization, and location. Given a sequence of tokens $(x_1, \dots, x_N)$, a NER system has to output triples (t_s, t_e, k) where $t_s, t_e \in \{1, \dots, N\}$ are the start and end token indices of an entity, respectively, and k is a named entity class. We cast NER as a sequence labeling task that performs unified entity identification and classification using the IOB2 tagging scheme [51].

We use the Golden Collections of the First HAREM evaluation contests [43], First HAREM and MiniHAREM, as train and test sets, respectively, following previous works [6, 42]. Both datasets contain multidomain documents annotated with 10 NE classes: Person, Organization, Location, Value, Date, Title, Thing, Event, Abstraction, and Other. Table 2 show some dataset samples.

We employ the datasets on two distinct scenarios: a Total scenario that considers all 10 classes, and a Selective scenario that includes only 5 classes (Person, Organization, Location, Value, and Date). Table 3 presents some dataset statistics. We set aside 7% of First HAREM documents as a holdout validation set.

To perform NER, we experiment with two architectures. In the simplest architecture, we attach a linear classifier layer on top of BERTimbau to predict the tag of each token independently. The model is trained using cross-entropy loss. We compute predictions and losses only for the first wordpiece of each token, ignoring word continuations, as depicted in Figure 1.

Since Linear-Chain Conditional Random Fields (CRF) [21] is widely adopted to enforce sequential classification in sequence labeling tasks [6, 22, 42], we also experiment with employing a CRF layer after the linear layer. We refer readers to Lample et al. [22] for a detailed explanation on CRF loss formulation and decoding procedure.

NER performance is evaluated using CoNLL 2003 [52] evaluation script,⁶ that computes entity-level precision, recall, and micro F1-score on exact matches. In other words, precision is the percentage of named entities predicted by the model that are correct, recall is the percentage of corpus entities that were correctly predicted and F1-score is the harmonic mean of precision and recall.

⁶<https://www.clips.uantwerpen.be/conll2002/ner/bin/conllevall.txt>

Table 1

Five samples of ASSIN2 dataset. Each sample is composed of a sentence pair and its gold STS relatedness score (a contiguous value from 1 to 5) and RTE label (Entailment or None). Actual dataset examples are in Portuguese, with English translations provided by us.

Gold STS/RTE	Sentence pair
5.0 / Entailment	A: <i>Os meninos estão de pé na frente do carro, que está queimando.</i> B: <i>Os meninos estão de pé na frente do carro em chamas.</i>
	English translation: A: The boys are standing in front of the car, which is burning. B: The boys are standing in front of the burning car.
4.0 / Entailment	A: <i>O campo verde para corrida de cavalos está completamente cheio de jockeys.</i> B: <i>Os jockeys estão correndo a cavalos no campo, que é completamente verde.</i>
	English translation: A: The green field for horse races is completely full of Jockeys. B: The Jockeys are racing horses on the field, which is completely green.
3.0 / Entailment	A: <i>A gruta com interior rosa está sendo escalada por quatro crianças do Oriente Médio, três meninas e um menino.</i> B: <i>Um grupo de crianças está brincando em uma estrutura colorida.</i>
	English translation: A: Four middle eastern children, three girls and one boy, are climbing on the grotto with a pink interior. B: A group of kids is playing in a colorful structure.
2.0 / None	A: <i>Não tem nenhuma pessoa descascando uma batata.</i> B: <i>Uma pessoa está fritando alguma comida.</i>
	English translation: A: There is no one peeling a potato. B: A person is frying some food.
1.0 / None	A: <i>Um cachorro está correndo no chão.</i> B: <i>A menina está batucando suas unhas.</i>
	English translation: A: A dog is running on the ground. B: The girl is tapping her fingernails.

4.3. Document context and max context evaluation for token-level tasks

In token-level tasks such as NER, we use document context for input examples instead of sentence context to take advantage of longer contexts when encoding token representations from BERT. Following the approach of Devlin et al. [11] on the SQuAD dataset, examples longer than S tokens are broken into spans of length up to S using a stride of D tokens. Each span is used as a separate example during training. During evaluation, however, a single token T_i can be present in $W = \frac{S}{D}$ multiple spans s_j , and so may have up to W distinct predictions $y_{i,j}$. Each token's final prediction is taken from the span where the token is closer to the central position, that is, the span where it has the most contextual information. Figure 1 illustrates this procedure.

5. Experiments

In this section, we present the experimental setup and results for BERT pretrainings and evaluation tasks. We conduct additional experiments to explore the usage of BERTimbau as a fixed extractor of contextual embeddings and also assess the impact of the long pretraining stage.

Table 2

FirstHAREM dataset samples. Gold named entities are enclosed by brackets with subscripted labels. Actual dataset examples are in Portuguese, with English translations provided by us.

A onça, ou jaguar, é um mamífero ([Panthera] _{THING} onça), da ordem dos carnívoros, família dos felídeos, encontrado em todo o continente americano, dos [EUA] _{LOC} à [Argentina] _{LOC} e em todo o [Brasil] _{LOC} .
English translation: The jaguar is a mammal ([Panthera] _{THING} onça), of the order of carnivores, family of felids, found throughout the American continent, from the [USA] _{LOC} to [Argentina] _{LOC} and throughout [Brazil] _{LOC} .
[Almeida Henriques] _{PER} ([A.H.] _{PER}): O [CEC] _{ORG} foi criado numa lógica de unir as associações da [Região Centro] _{LOC} , quer sejam industriais, quer sejam comerciais, quer sejam agrícolas.
English translation: [Almeida Henriques] _{PER} ([A.H.] _{PER}): The [CEC] _{ORG} was created in a logic of uniting the associations of the Center Region, whether industrial, commercial or agricultural.
Entre os mais importantes destacam-se o de [Shanta Durga] _{TITLE} e o de [Shri Munguesh] _{TITLE} , construídos há [400 anos] _{VALUE} .
English translation: Among the most important are [Shanta Durga] _{TITLE} and [Shri Munguesh] _{TITLE} , built [400 years] _{VALUE} ago.
Para aqueles que vão participar do processo seletivo, o professor de [Direito Previdenciário] _{ABS} [Fábio Zambite] _{PER} dá uma dica importante: os candidatos devem estudar com bastante atenção o [Decreto 3.048/99] _{TITLE} , que aprova o [Regulamento da Previdência Social] _{TITLE} .
English translation: For those who are going to participate in the selection process, Professor of [Social Security Law] _{ABS} [Fábio Zambite] _{PER} gives an important tip: candidates must carefully study [Decree 3.048/99] _{TITLE} , which approves the [Social Security Regulation] _{TITLE} .
[A Mulher no Início do Novo Século] _{EVENT} Dia [15 de Maio] _{TIME} , pelas [9.30H] _{TIME} , no [Cine-Teatro Caridade] _{LOC} , em [Moura] _{LOC} irá realizar-se um Fórum intitulado [A Mulher no Início do Novo Século] _{ABS} , tendo como organização a [Câmara Municipal de Moura] _{ORG} e a colaboração da [Associação de Mulheres do Concelho de Moura] _{ORG} .
English translation: [Women at the Beginning of the New Century] _{EVENT} On the [15th of May] _{TIME} , at [9.30 am] _{TIME} , at the [Cine-Teatro Caridade] _{LOC} , in [Moura] _{LOC} , a Forum entitled [Women at the Beginning of the New Century] _{ABS} will take place, organized by the [Moura City Council] _{ORG} with the collaboration of the [Moura's Women Association Board] _{ORG} .
[Touro] _{OTHER} é o signo seguinte. O sol o visita entre [21 de abril] _{TIME} e [21 de maio] _{TIME} , domicílio de [Vênus] _{THING} . English translation: [Taurus] _{OTHER} is the next sign. The sun visits him between [April 21] _{TIME} and [May 21] _{TIME} , home of [Venus] _{THING} .

5.1. Pretrainings

Following the original BERT work [11], we train BERTimbau models of two sizes: Base ($L = 12$ layers, $H = 768$, 12 attention heads, and 110M parameters) and Large ($L = 24$ layers, $H = 1024$, 16 attention heads and 330M parameters). The maximum sentence length is set to $S = 512$ tokens. We train cased models only since we focus on general purpose models and capitalization is relevant for tasks like named entity recognition [6, 11].

The models are pretrained for 1,000,000 steps. We use a learning rate of $1e-4$, learning rate warmup over the first 10,000 steps followed by a linear decay of the learning rate over the remaining steps.

For BERTimbau Base models, the weights are initialized with the checkpoint of Multilingual BERT Base (discarding the word embeddings and the MLM head weights that are from a different vocabulary). We use a batch size of 128 and sequences of 512 tokens the entire training. This training takes 4 days on a TPU v3-8 instance and performs about 8 epochs over the pretraining data.

For BERTimbau Large, the weights are initialized with the checkpoint of English BERT Large (again discarding the word embeddings and MLM head weights that are from a different vocabulary), because Multilingual BERT is

Table 3

Dataset statistics for the HAREM I corpora. The Tokens column refers to whitespace and punctuation tokenization.

Dataset	Documents	Tokens	Entities in scenario	
			Selective	Total
First HAREM	129	95585	4151	5017
MiniHAREM	128	64853	3018	3642

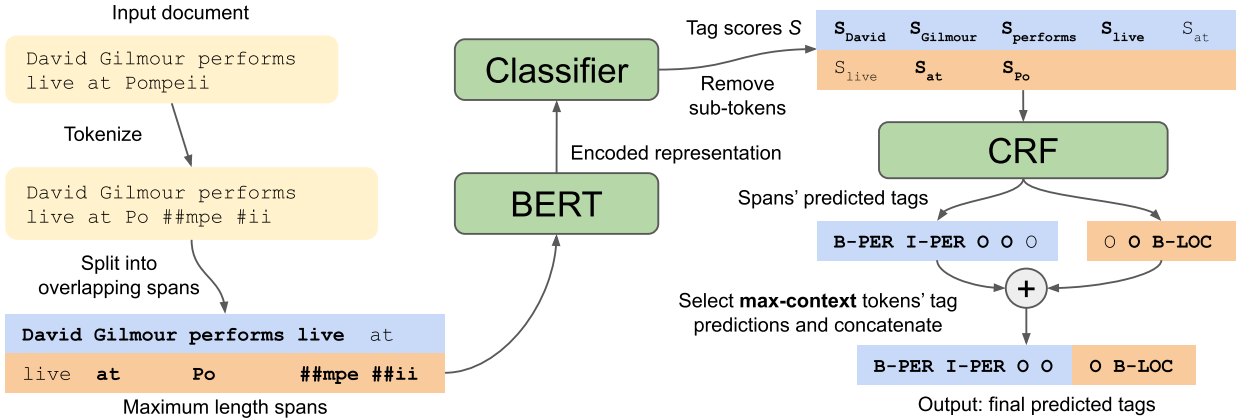


Figure 1: Illustration of the proposed method for the NER task described in 4.3. Given an input document, the text is tokenized using WordPiece [56] and the tokenized document is split into overlapping spans of the maximum length using a fixed stride (= 3, in the example). Maximum context tokens of each span are marked in bold. The spans are fed into BERT and then into the classification model, producing a sequence of tag scores for each span. The scores of subtoken entries (starting with ##) are removed from the spans and the remaining tags scores are passed to the CRF layer — if it is employed, otherwise the highest tag scores are used independently. The maximum context tokens are selected and concatenated to form the final predicted tags.

only available at Base size. Since it is a bigger model with longer training time, we follow the instructions of Devlin et al. [11] and use sequences of 128 tokens in batches of size 256 for the first 900,000 steps and then sequences of 512 tokens and batch size 128 for the last 100,000 steps. This training takes 7 days on a TPU v3-8 instance and performs about 6 epochs over the training data.

Training loss curves for both pretrainings are shown in Figure 2. It can be seen that there is a sharp decrease in loss over the initial 100k steps, which can be interpreted as the models learning the word embeddings that are initialized randomly. The losses slowly decrease afterwards until the end of the training. A steep decrease in BERTimbau Large loss can be noticed in step 900,000, which marks the beginning of the pretraining using sequences of 512 tokens.

Note that in the calculation of the number of epochs, we are taking into consideration a duplication factor of 10 when generating the input examples. This means that under 10 epochs, the same sentence is seen with different masking and sentence pair in each epoch, which is effectively equal to dynamic example generation proposed by RoBERTa [25].

5.2. Fine-tunings on evaluation tasks

For all evaluation experiments, we fine-tune the complete model and we use a learning rate schedule of warmup over the first 10% steps followed by linear decay of the learning rate over the remaining steps. Similar to pretraining, we use BERT’s AdamW optimizer implementation with $\beta_1 = 0.9$, $\beta_2 = 0.999$ and L2 weight decay of 0.01. We perform early stopping and select the best model on the validation set of each dataset.

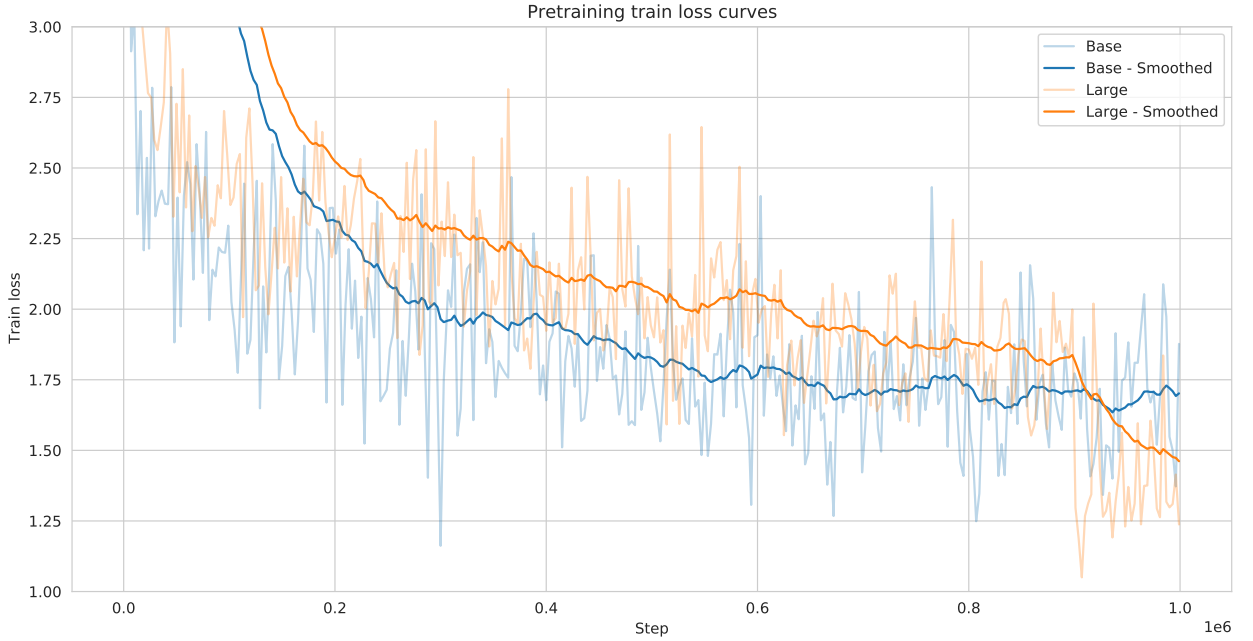


Figure 2: Training loss curves for BERTimbau Base and BERTimbau Large pretrainings. Smoothed curves are exponential moving averages with smoothing factor $\alpha = 0.95$. A sharp decrease in BERTimbau Large training loss can be noticed after step 900,000, when training begins using sequences of 512 tokens.

Table 4

Test scores for STS and RTE tasks on ASSIN2 dataset. We compare our models to the best published results. Best scores in bold. Reported values are the average of multiple runs (standard deviation in parenthesis) with different random seeds. Star ($\star$) denotes primary metrics. $\dagger$: ensemble technique. $\ddagger$: extra training data.

Row	Model	STS		RTE	
		Pearson ($\star$)	MSE	F1 ($\star$)	Accuracy
1	mBERT + RoBERTa-Large-en (Averaging) [40] $\dagger$	0.83	0.91	84	84.8
2	mBERT + RoBERTa-Large-en (Stacking) [40] $\dagger$	0.785	0.59	88.3	88.3
3	mBERT (STS) and mBERT-PT (RTE) [39] $\ddagger$	0.826	0.52	87.6	87.6
4	USE+Features (STS) and mBERT+Features (RTE) [13]	0.800	0.39	86.6	86.6
5	mBERT+Features [13]	0.817	0.47	86.6	86.6
6	mBERT (ours)	0.809 (0.004)	0.58 (0.04)	86.8 (0.4)	86.8 (0.4)
7	BERTimbau Base	0.836 (0.007)	0.58 (0.03)	89.2 (0.8)	89.2 (0.8)
8	BERTimbau Large	0.852 (0.003)	0.50 (0.03)	90.0 (0.4)	90.0 (0.4)

5.3. Sentence Textual Similarity and Recognizing Textual Entailment results

For this experiment, we train BERTimbau Base with learning rate of $4e-5$ and batch size 32 for 10 epochs, and BERTimbau Large with learning rate of $1e-5$, batch size 8 for 5 epochs. We also train mBERT to compare it to BERTimbau models. mBERT is trained using learning rate of $1e-5$ and batch size 8 for 10 epochs.

5.3.1. Results

Our results for both tasks are shown in Table 4. We compare our results to the best-performing submissions to official ASSIN2. All compared works employ mBERT or a Transformer-based architecture in their approaches. In the following paragraphs, we refer to each work using their corresponding row numbers in Table 4.

Table 5

Results of NER task (Precision, Recall and micro F1-score) on the test set (MiniHAREM). Best results in bold. Reported values are the average of multiple runs (standard deviation in parenthesis) with different random seeds. Star (★) denotes primary metrics.

Row	Architecture	Total scenario			Selective scenario		
		Prec.	Rec.	F1 (★)	Prec.	Rec.	F1 (★)
1	CharWNN [42]	67.2	63.7	65.4	74.0	68.7	71.2
2	LSTM-CRF [6]	72.8	68.0	70.3	78.3	74.4	76.3
3	BiLSTM-CRF+FlairBBP [44]	74.9	74.4	74.6	83.4	81.2	82.3
4	mBERT	71.6 (1.1)	72.7 (0.4)	72.2 (0.6)	77.0 (0.8)	78.8 (0.7)	77.9 (0.6)
5	mBERT + CRF	74.1 (0.7)	72.2 (0.8)	73.1 (0.7)	80.1 (0.4)	78.3 (0.6)	79.2 (0.2)
6	BERTimbau Base	76.8 (0.8)	77.1 (0.7)	77.2 (0.7)	81.9 (0.7)	82.7 (0.4)	82.2 (0.4)
7	BERTimbau Base + CRF	78.5 (0.8)	76.8 (1.1)	77.6 (0.8)	84.6 (1.2)	81.6 (1.0)	83.1 (1.1)
8	BERTimbau Large	77.9 (0.4)	78.0 (0.3)	77.9 (0.3)	81.3 (0.9)	82.2 (0.9)	81.7 (0.7)
9	BERTimbau Large + CRF	79.6 (0.8)	77.4 (1.0)	78.5 (0.8)	84.9 (0.9)	82.5 (0.6)	83.7 (0.2)

BERTimbau models achieve the best results on the primary metrics of both STS and RTE tasks, with the large model performing significantly better than the base variant. The previous highest scores (rows 1 and 2) for both STS Pearson’s correlation and RTE F1 score are from ensemble techniques that combine mBERT fine-tuned on original ASSIN2 data and an English RoBERTa-Large fine-tuned on ASSIN2 data automatically translated to English. The averaging ensemble uses 2 models and the stacking ensemble uses 10 distinct fine-tuned models — 5-fold stacking which results in 5 mBERT and 5 RoBERTa trained models. While this approach shows an interesting application of English models to Portuguese tasks, our BERTimbau models achieve higher performance using a single model and, hence, demand lower compute resources in both fine-tuning and inference stages.

Regarding our implementation using mBERT (row 6), it presents a lower performance compared to BERTimbau models, which highlights the benefits of Portuguese pretraining of BERTimbau. For STS task, we note that mBERT achieves the same MSE as BERTimbau Base, even though Pearson correlation is lower. Comparing it to other works’ approaches, better performances are achieved using extra supervised training data and further pretraining of mBERT on Portuguese data (row 3), and also by combining it with hand-designed features (rows 4 and 5).

5.4. NER task

In this section, we refer to the 2 architectures defined in Section 4.2 as BERT and BERT-CRF. Long examples are broken into spans using a stride of $D = 128$ as explained in Section 4.3.

The model parameters are divided in two groups with different learning rates: $5e-5$ for BERT model and $1e-3$ for the classifier. We train the models for up to 50 epochs using a batch size of 16. Models with CRF are trained for up to 15 epochs.

In addition to BERTimbau Base and Large, we also train mBERT to compare monolingual versus multilingual model performances. mBERT is fine-tuned with the same hyperparameters.

It is common in NER for the vast majority of tokens not to belong to named entities (and have tag label “O”). To deal with this class imbalance, we initialize the classifier’s bias term of the “O” tag with a value of 6 in order to promote a better stability in early training [24]. We also use a weight of 0.01 for “O” tag losses.

When evaluating, we produce valid predictions by removing all invalid tag transitions for the IOB2 scheme, such as “I-” tags coming directly after “O” tags or after an “I-” tag of a different class. This post-processing step trades off recall for a possibly higher precision.

5.4.1. Results

The main results of our NER experiments are presented in Table 5. We compare the performances of our models on the two scenarios (total and selective) defined in Section 4.2 to results of previous works. The models of rows 1 to 3 show the progress of neural network approaches for this dataset over the recent years. The previous best result (row 3), achieved by BiLSTM-CRF+FlairBBP model, uses Portuguese Flair Embeddings, which are contextual embeddings

Table 6

NER performances (Precision, Recall and F1-score) on the test set (MiniHAREM) using BERTimbau as contextual embeddings in a feature-based approach. Star (★) denotes primary metrics.

Architecture	Total scenario			Selective scenario		
	Prec.	Rec.	F1 (★)	Prec.	Rec.	F1 (★)
mBERT + LSTM-CRF	74.7	69.7	72.1	80.6	75.0	77.7
BERTimbau Base + LSTM-CRF	78.3	73.2	75.6	84.5	78.7	81.6
BERTimbau Large + LSTM-CRF	77.4	72.4	74.8	83.0	77.8	80.3

extracted from character-level language models [1].

Our best model, BERTimbau Large + CRF (row 9), outperforms the best published results improving the F1-score by 3.9 points on the total scenario and by 1.4 point on the selective scenario. Interestingly, Flair embeddings outperform BERT models on English NER [1, 11].

There is a large performance gap between BERTimbau and mBERT, which reinforces the advantages of monolingual models pretrained on multidomain data compared to mBERT, that is trained only on Wikipedia articles. This result is on par with other monolingual BERT works.

The CRF layer consistently brings performance improvements in F1 in all settings. However, F1 increases are pushed by a large boost in precision that is often associated with lower recall. It is worth noting that, without CRF, BERTimbau Large shows a close but inferior performance to the Base variant on the selective scenario. This result suggests that a more controlled fine-tuning scheme might be required in some cases, such as partial layer unfreezing or discriminative fine-tuning [33] — usage of lower learning rates for lower layers —, given that it is a higher capacity model trained on few data.

5.5. BERTimbau as contextual embeddings

In this experiment, we evaluate BERTimbau as a fixed extractor of contextual embeddings that we use as input features to train a downstream model on the NER task. In other words, BERTimbau’s weights are kept frozen and only the downstream model’s weights are optimized during training, in contrast to the fine-tuning experiments where the complete model is optimized. This setup can be interesting in lower resource scenarios in which several tasks are to be performed on the same input text: the extraction of contextual embeddings —which is the most expensive stage, —can be computed once and then shared across several smaller task-specific models.

In this feature-based approach, we train a BiLSTM-CRF model with 1 layer and 100 hidden units followed by a linear classifier layer for up to 50 epochs. Instead of using only the hidden representation of BERT’s last encoder layer, the representation $\mathbf{T}_i$ of each token is taken as the sum its representations of the last 4 layers, as proposed by the original work [11]:

$$\mathbf{T}_i^{\text{fixed}} = \mathbf{T}_i^{(L)} + \mathbf{T}_i^{(L-1)} + \mathbf{T}_i^{(L-2)} + \mathbf{T}_i^{(L-3)}, \quad (7)$$

where $\mathbf{T}_i^{(j)} \in \mathbb{R}^H$ is the output of the j -th Transformer Encoder layer at the i -th position and L is the number of layers. The resulting architecture resembles the BiLSTM-CRF model [22] but using BERT contextual embeddings instead of fixed word embeddings.

5.5.1. Results

We present the results on Table 6. Models of the feature-based approach perform significantly worse compared to the ones of the fine-tuning approach. The performance gap is found to be much higher than the reported values for NER on English language [11, 33] and reaches up to 2 points on BERTimbau Base and 3.5 points on BERTimbau-Large, although it can probably be reduced by further hyperparameter tuning.

In this setup, BERTimbau Base+BiLSTM-CRF achieves similar performances to BiLSTM-CRF+FlairBBP (row 3 of Table 5), which also uses contextual embeddings and a similar architecture. BERTimbau shows a slightly lower F1-score in the Selective scenario but higher F1-score in the Total scenario.

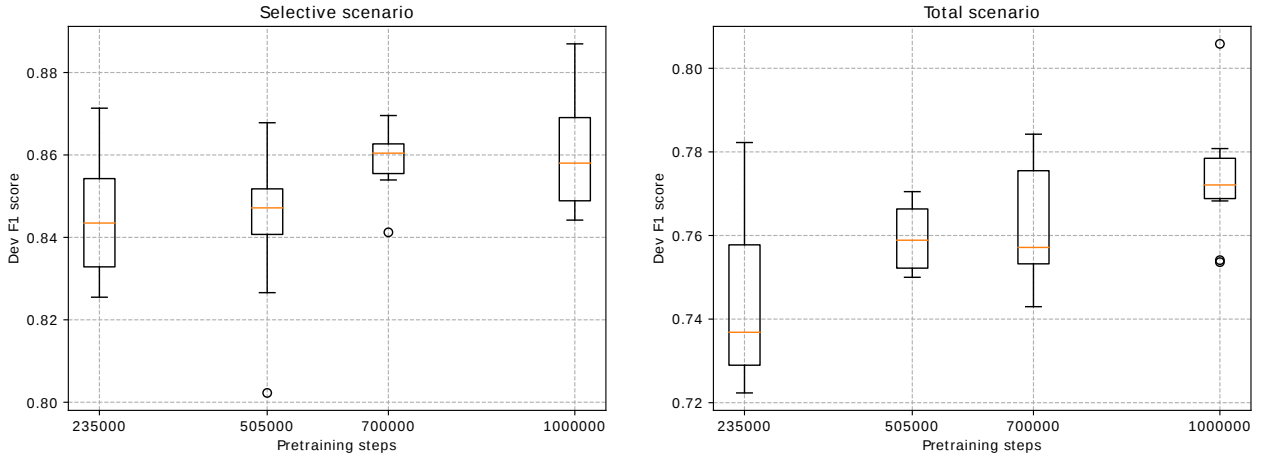


Figure 3: Performance of BERTimbau Base on NER task using intermediate checkpoints of the pretraining stage. Scores reported on the validation set.

It is worth mentioning that BERTimbau models in this feature-based approach achieve better performances than a fine-tuned mBERT on this same task. While BERTimbau Large is the highest performer when fine-tuned, we observe that it experiences performance degradation when used in this feature-based approach, performing worse than the smaller Base variant but still better than mBERT.

5.6. Impact of pretraining steps

To assess the impact of the number of pretraining steps on the performance of downstream tasks, we repeat part of the NER fine-tuning experiment (Section 5.4) using intermediate checkpoints of BERTimbau Base pretraining procedure. We train BERT models (without CRF) using the checkpoints of steps 235k, 505k and 700k, which correspond to 23.5%, 50.5% and 70% of the complete pretraining of 1000k steps, respectively. All models are trained with the same hyperparameters and experimental setup described in Section 5.4.

The results are shown in Figure 3. Performances on the downstream task increase non-linearly with pretraining steps, with diminishing returns as pretraining progresses. This is an expected result, as test performance of pretraining tasks are shown to follow a power law on the number of pretraining steps [17].

6. Tokenization analysis

One possible advantage of a monolingual BERT over multilingual BERT can be related to the WordPiece tokenizer vocabulary. The vocabulary size is a hyperparameter that limits the number of distinct recognizable tokens, which affects the size of the input token embedding matrix. Most monolingual BERT models have vocabulary sizes in the range of 30,000 to 50,000 tokens [11, 25, 23, 4]. In comparison, mBERT has a vocabulary of 120,000 tokens, which has to encompass tokens of over 100 languages and a variety of alphabets. When considering the usage of mBERT on a single specific language, the effective vocabulary size is usually much smaller than a monolingual equivalent, resulting in longer tokenized sequences. This happens because, in smaller vocabularies generated by BPE, only very frequent words will be present as individual tokens, causing the tokenization of most words to be composed of multiple subword units.

Considering that dot-product attention layers have quadratic complexity that imposes limitations on input sequence size of BERT and Transformer models in general [53], a more efficient tokenization that produces shorter sequences allows inputting larger textual context in a sequence of maximum length S . This limitation is often encountered in sequence-level tasks such as document classification of long documents [50].

One can also hypothesize that a tokenization that often breaks words into multiple subword units imposes a harder task on the model, since instead of receiving an embedding vector that readily represents the original word, the model will receive several vectors — one vector for each subword unit — that will have to be combined inside the model to form a complete word representation.

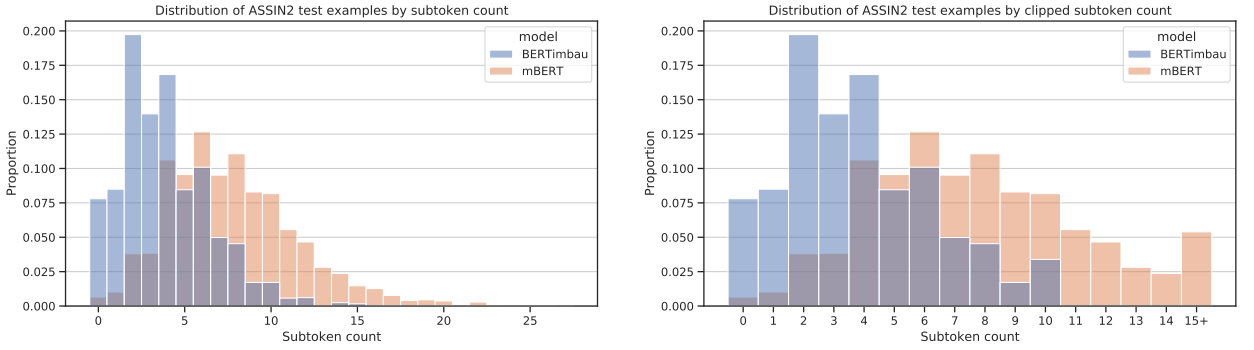


Figure 4: Distribution of ASSIN2 test set examples binned by subtoken count in the tokenization of the concatenated premise and hypothesis texts, for BERTimbau and mBERT (left-side). A subtoken is any word continuation token that starts with “##”. The bin at $x = 0$ contains examples whose premise and hypothesis tokenizations are composed of only whole words. The histogram on the right-side is a clipped version that aggregates the right tail of the distributions in the $x = 10$ and $x = 15$ bins for BERTimbau and mBERT, respectively.

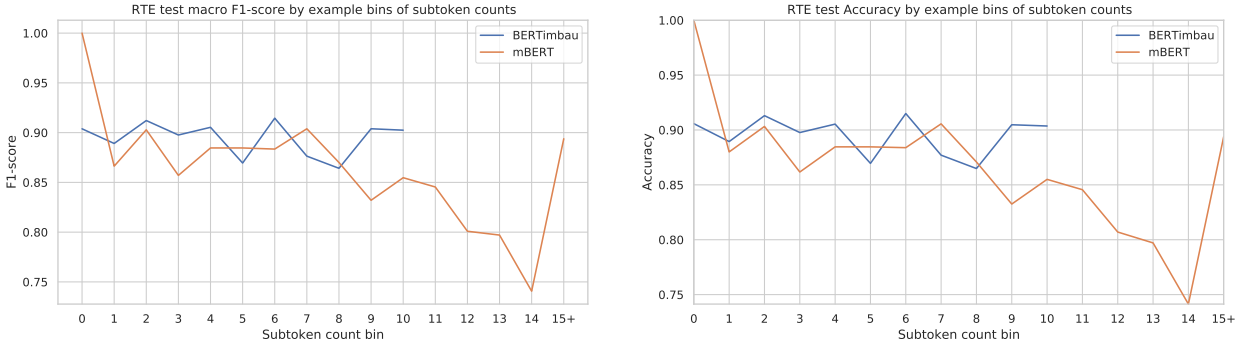


Figure 5: Metrics of RTE task on ASSIN2 test set examples computed separately for each bin of the distribution of the right side of Figure 4.

In this experiment, we analyze the tokenizations produced by BERTimbau’s and mBERT’s tokenizers and compare the produced tokenized sequences for the evaluation tasks’ datasets. We compare the sequence lengths for each downstream task dataset and assess how subword unit tokenization may affect the performance of each task.

6.1. Tokenization effects on Sentence Textual Similarity and Recognizing Textual Entailment tasks

To investigate the effects of the tokenization on the RTE and STS tasks, we tokenize the ASSIN2 test set examples using BERTimbau and mBERT tokenizers. The examples are then binned by the subtoken count in the premise and hypothesis texts’ tokenizations, as can be seen in Figure 4. The mBERT tokenizer produces a median of 7 subtokens per example, while BERTimbau has a distribution skewed to the lower end and a median value of 3.5. Less than 1% of the examples tokenized using the multilingual vocabulary are composed of only whole words, while this proportion is 7.5% for the Portuguese tokenizer. Since both distributions have a right tail of low proportion bins, the right tails are clipped to the bins $x = 10$ for BERTimbau and $x = 15$ for mBERT, as shown in the right side of Figure 4, and these bins are used for the following metrics analysis.

For each example bin of the distribution, we compute the evaluation metrics for RTE and STS tasks to see how performance vary as tokenizations break words into more pieces, as shown in Figures 5 and 6. For the RTE task, there is almost no variation of F1-score and accuracy for BERTimbau as subtoken count increases. For mBERT, it appears to have a performance degradation beginning at the 9 subtokens bin, even though the 15+ bin recovers the performance of the lower bins. It is noticeable that mBERT performs on par with BERTimbau in the lower subtoken bins, and its

BERT for Brazilian Portuguese

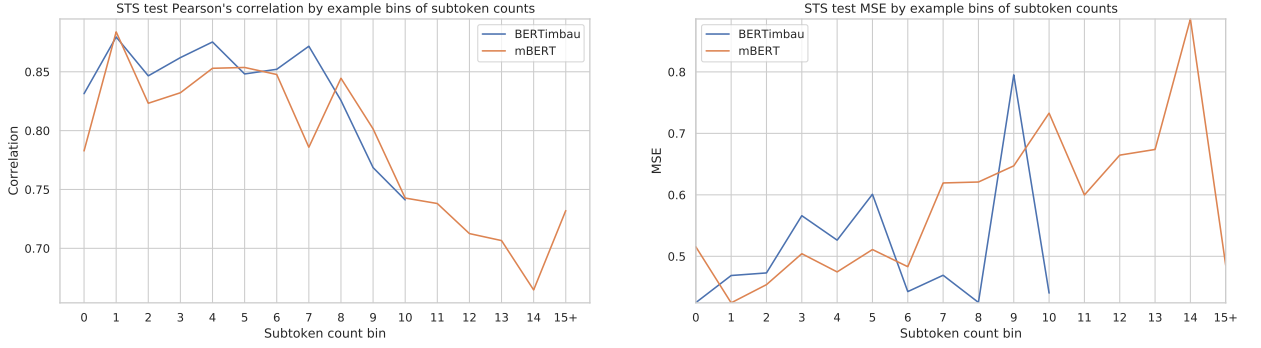


Figure 6: Metrics of STS task on ASSIN2 test set examples computed separately for each bin of the distribution of the right side of Figure 4. Computation of Pearson's correlation uses global mean values for ground-truth and prediction similarity scores.

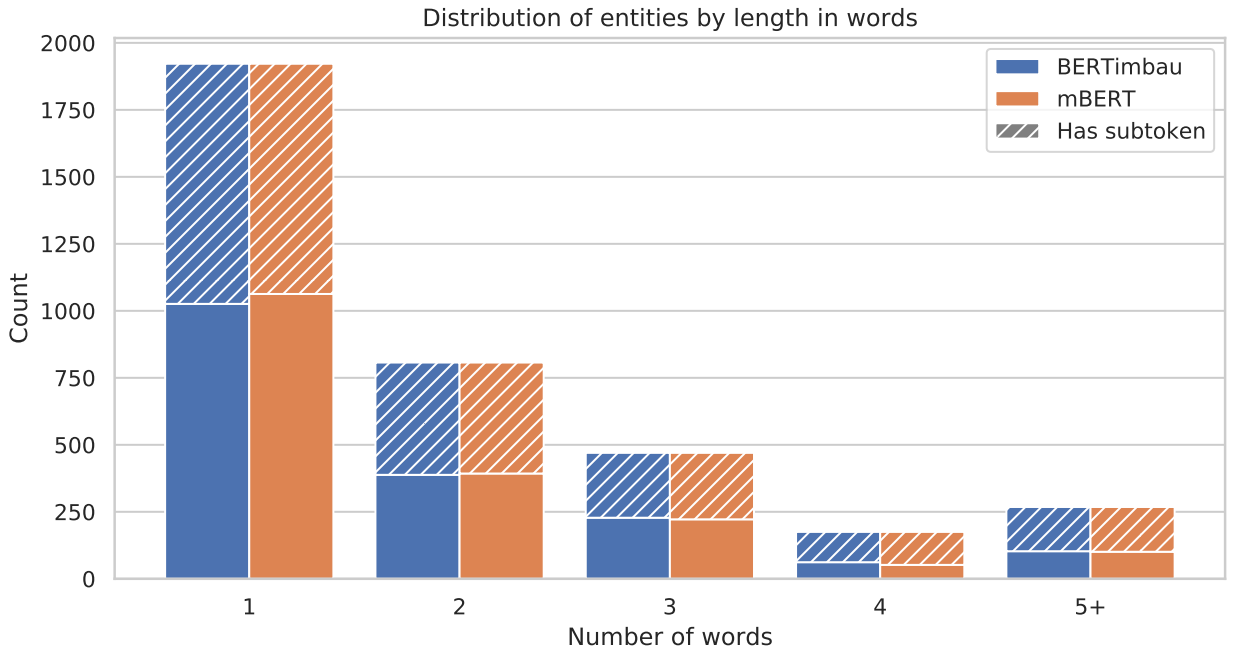


Figure 7: Distribution of ground-truth entities of Mini HAREM dataset by number of words and presence of subtoken (word continuation token that starts with “##”), for BERTimbau and mBERT.

global metrics are affected by the higher bins with worse performance, that comprise over 30% of the test set. Similar conclusions can be drawn for the STS task metrics, with BERTimbau metrics showing less overall variation while mBERT shows degradation on the higher bins. We argue, however, that it is not possible to draw precise conclusions, since bins at the distribution tails may have as few as 30 to 100 examples and, as such, the metrics of these bins can be dominated by the presence of easy or hard examples or with rarer words.

6.2. Tokenization effects on Named Entity Recognition

Given that Named Entity Recognition is a token-level task and the metrics are computed on entity-level, we take a distinct approach and analyze how the tokenization of the entities' words may affect the model performance. This way, we compare the tokenizations using BERTimbau and mBERT vocabularies on the ground-truth and predicted entities of the test dataset, Mini HAREM. The compared models are the BERT-CRF architecture on the Total scenario.

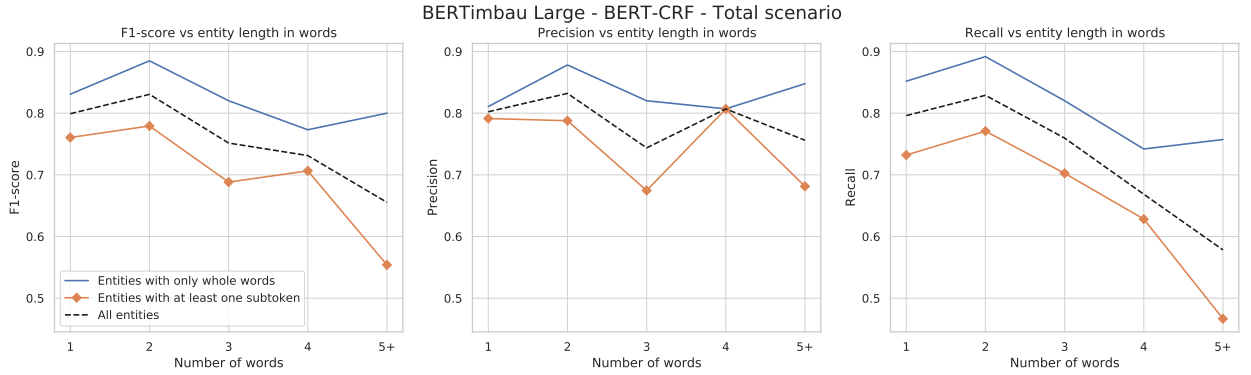


Figure 8: Metrics of BERTimbau Large (BERT-CRF) on Named Entity Recognition task on the test dataset (Mini HAREM, Total scenario) binned by entity word count.

When casting NER as a sequence tagging problem, it is intuitively expected that longer entities might be harder to predict accurately. This comes from the fact that an entity composed of more words is encoded as a longer tag sequence and any incorrectly predicted tag yields a wrong entity prediction, which hurts both recall and precision. Considering this proposition, we bin the ground-truth and predicted entities by word count and, inside each bin, we distinguish between entities whose tokenization contains only whole words or contains at least one subtoken, as shown in Figure 7 for the ground-truth entities. It is worth emphasizing the definition of word in this context: we consider as words any sequence of characters that are produced by splitting a text into whitespace and punctuation characters, considering each punctuation character as a separate word. The length of an entity in words is independent of the vocabulary. Each word is then tokenized into one or multiple subword units by WordPiece tokenization using the model vocabulary, affecting the presence of subtoken or not.

As can be seen in Figure 7, the proportion of entities that have subtokens inside each bin is very similar between BERTimbau and mBERT vocabularies. This is not unexpected, since it is a general NER dataset and entities often contain proper names which are commonly rarer words. Even though the Portuguese vocabulary contains a larger set of common Portuguese proper names than the multilingual vocabulary, the opposite holds for foreign proper names, for instance, and these effects roughly balance each other out in this case.

We separately compute NER metrics (F1-score, Precision and Recall) for both analyzed models for each entity bin of Figure 7, as shown for BERTimbau in Figure 8 and for mBERT in Figure 9. By observing the dashed lines of the F1-score and recall plots, it can be seen that the proposition of longer entities showing worse performances hold for both models. The most relevant finding is that both models show worse performances to detect entities that have at least one subtoken, across all entity lengths, compared to entities that are composed of only whole words. The degree of performance degradation for increasing entity length is also higher for entities that contain subtokens.

We did not expect the presence of subtokens inside the entities to impact the task performance in such a strong and consistent manner. When defining the NER task, it is often stated that the class of a named entity is dependent not only on the entity itself, but also heavily dependent on its surrounding words [57]. For instance, consider the act of replacing a person's or organization's name in a sentence by another one of the same class. This replacement should not make the entity necessarily easier or harder to detect — as long as the meaning and syntax of the sentence are preserved in this process —, since the surrounding context might give enough information to infer the entity class nonetheless. However, this observed performance degradation suggests that the choice of the replacement entity might affect the model's capabilities if it contains subtokens or not.

6.3. Discussion

Our experiments reveal a correlation between word segmentation into multiple subwords and task performance, with a higher degradation in the case of mBERT. Analyzing the impacts of the tokenization on these models can be an area of further research that has not been much explored. SciBERT [3] — a BERT trained on scientific articles in English — shows that having an in-domain vocabulary is beneficial, but they argue that larger benefits come rather from fine-tuning BERT on in-domain data than from a more suited vocabulary. These two factors are present in BERTimbau, since it is trained on more Portuguese data (and more diverse) than mBERT.

BERT for Brazilian Portuguese

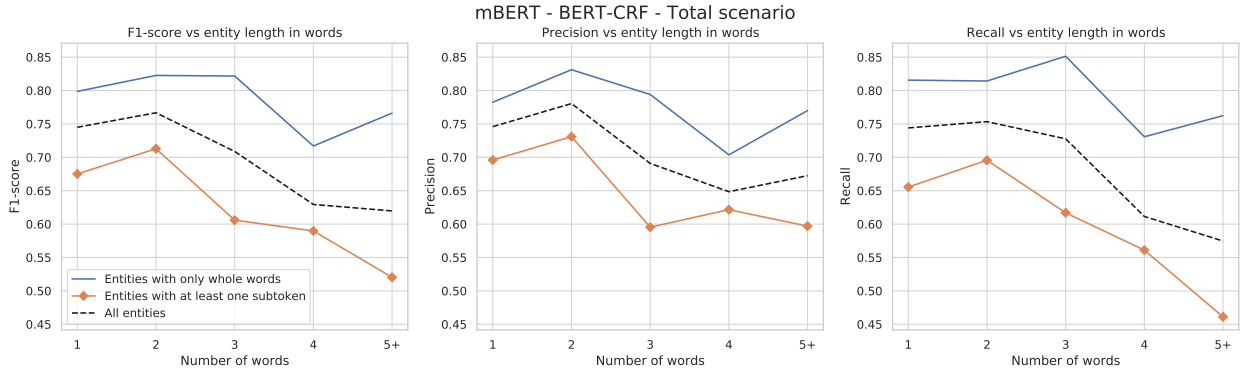


Figure 9: Metrics of mBERT (BERT-CRF) on Named Entity Recognition task on the test dataset (Mini HAREM, Total scenario) binned by entity word count.

However, we believe there are other factors that, along with the tokenization method, might be hurting the performance of these models and can be subjects of research. While subword unit tokenization is a more robust alternative to word-level tokenization, allowing the representation of out-of-vocabulary words and using much smaller vocabularies, there is performance degradation when tokenization deviates from the ideal scenario where words are kept intact. We hypothesize that BERT architecture is not being capable of easily reconstructing a high quality word representation from multiple independent subword units. Future works could experiment with techniques such as subword regularization [18] as to stimulate the learning of compositionality of multiple subword units.

7. Conclusion

In this work, we advance the study of deep learning models for NLP in Portuguese, especially the usage of pretrained language models in a transfer learning approach. We train BERT models for Brazilian Portuguese and evaluate their performances on three downstream NLP tasks.

In the pretraining stage, we use Wikipedia articles to generate a Portuguese vocabulary and then leverage millions of webpages from the brWaC corpus as unlabeled data to train Portuguese BERT models on self-supervised objectives. On the evaluation stage, we fine-tune the models supervisely on downstream tasks in two distinct experiments.

In the first experiment, we fine-tune our BERTimbau models on the ASSIN2 dataset to jointly solve Sentence Textual Similarity (STS) and Recognizing Textual Entailment (RTE) tasks. BERTimbau achieves state-of-the-art performances in both tasks, surpassing Multilingual BERT (mBERT) and previously published results in the literature, that comprise both Portuguese specific models and multilingual approaches.

In the second experiment, we fine-tune BERTimbau on named entity recognition (NER) task using the FirstHAREM and MiniHAREM datasets. We experiment with two NER architectures: plain BERT and BERT-CRF. Again, our best model achieves state-of-the-art results and shows a large performance improvement over mBERT and the previously best published result, that uses Portuguese Flair embeddings in a contextual embeddings setup, especially in the hardest Total scenario that considers all 10 named entity classes.

In the three tasks, BERTimbau Large shows a performance slightly higher than BERTimbau Base, as expected of a higher capacity model pretrained and fine-tuned to small labeled datasets. Hence, the choice of which model size to use depends primarily on the trade-off between performance and compute resources, especially during inference usage.

In additional experiments, we assess the usage of BERTimbau in a contextual embeddings setup by freezing its weights and training BERTimbau-BiLSTM-CRF models on NER task. Even though there is a notable performance drop, we show that contextual embeddings from BERTimbau Base outperform fine-tuned mBERT models, which can be a lower compute alternative for limited resource scenarios. We also validate the necessity of a long pretraining stage, that had been reported for English and other languages, for our Portuguese models by evaluating the performance of intermediate model checkpoints of the pretraining stage on NER task. Models pretrained for longer times show better performance in the end task, even though the pretraining stage had already started from pretrained checkpoints from mBERT and English BERT.

Lastly, we compare BERTimbau's Portuguese vocabulary to mBERT's multilingual vocabulary by looking at the produced tokenizations on the evaluation tasks. The Portuguese vocabulary produces smaller tokenized sentences, which corresponds to (1) keeping more words intact as a single token and (2) breaking words using a lower average number of subword units per word. We analyze how tokenizing words into multiple subword units might affect the model performance on the end tasks by binning task examples by tokenization statistics and computing evaluation metrics separately for each group. In particular to NER, we notice the models show inferior performance to detect named entities that contain at least one subword unit compared to only whole words. While this phenomenon could be related to the presence of rarer words in these examples, we hypothesize that the BERT architecture is not fully capable of reconstructing word representations from several subword units since it relies heavily on the positional embeddings that might not be sufficient. Further experiments and analyses can be performed to better understand these issues, such as exploring other evaluation tasks, vocabulary generation algorithms, regularization techniques or looking for alternatives to the simple positional embedding.

In regards to multilingual models, we notice mBERT is one of the first works in this area. There are more recent models, such as XLM [8] and XLM-R [7], that can be experimented with in future work. These models propose new training procedures that allow greater knowledge sharing across languages without sacrificing much per-language performance, and avoiding vocabulary dilution. Even though Portuguese is heavily present on the internet and, as such, it has enough unlabeled data to train large monolingual language models, the cross-lingual transfer allowed by multilingual models can be extremely beneficial for Portuguese by leveraging labeled datasets of other languages to alleviate the annotated data limitation that is commonly faced by NLP researchers and developers.

8. Acknowledgements

R Lotufo acknowledges the support of the Brazilian government through the CNPq Fellowship ref. 310828/2018-0. We would like to thank Google Cloud for research credits.

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